

■ PostureGuard AI

Real-Time Posture Analysis & Ergonomic Coach

Computer Vision Capstone Project · April 2026

1. Problem Statement

Poor posture during desk work is a global health crisis. Over **1 billion knowledge workers** spend 6–10 hours daily seated, with studies showing 60–80% develop significant musculoskeletal problems by age 40. Forward head posture, slouching, and shoulder imbalance cause chronic neck pain, back degeneration, headaches, and reduced lung capacity. Existing solutions — physiotherapy, ergonomic assessments, wearable posture correctors — are expensive, infrequent, or intrusive.

PostureGuard AI is a free, real-time, camera-based ergonomic coach that runs on any laptop and gives continuous, actionable posture feedback. No hardware, no subscription, no privacy risk.

2. Model & Approach

We use **MediaPipe BlazePose** (Google, pretrained) for real-time 33-landmark pose estimation at ~30 FPS on CPU — no GPU required. On top of the raw landmarks, we implement a custom **PostureAnalyzer** that computes four biomechanical metrics and derives a smoothed posture score.

Metric	Measurement	Threshold	Max Deduction
Forward Head	Ear-shoulder horizontal deviation (°)	>15°	35 pts
Shoulder Tilt	Left/right shoulder Y-asymmetry (%)	>8%	25 pts
Slouching	Hip-shoulder vertical deviation (°)	>20°	30 pts
Head Tilt	Eye-level asymmetry (%)	>5%	10 pts

Score formula: `score = max(0, 100 - Σ(weight × severity))` with EMA smoothing ($\alpha=0.15$) to prevent flickering.

3. Pipeline

Step	Component	Description
1	OpenCV	Capture 640×480 frame from webcam or video file
2	MediaPipe Pose	33 landmark prediction with per-joint visibility scores
3	PostureAnalyzer	Extract 4 metrics via angle/proportion geometry
4	EMA Filter	Smooth score over time ($\alpha=0.15$, 150ms effective window)
5	Flask API	Serialize JSON result to browser dashboard
6	Frontend	Render skeleton canvas, animated score ring, chart, alerts

4. Dataset

No custom training dataset is required — **MediaPipe BlazePose** generalizes across all adult body types and common seated postures out of the box. Threshold calibration was performed through:

- Manual testing with 5 subjects across 3 posture categories (good / moderate / poor)
- Cross-referencing with OSHA ergonomic guidelines and HSE UK seated workstation standards
- Iterative refinement to minimize false positives during natural movement (reaching, turning)

Self-annotated threshold values are stored as named constants in `inference.py` and can be tuned per-user. Sessions are logged to timestamped CSVs.

5. Challenges

Occlusion: Side-on angles hide one shoulder — resolved by using midpoint averaging between left/right landmarks with visibility weighting.

Noisy per-frame scores: EMA smoothing ($\alpha=0.15$) provides stable readout without lag — score responds within ~1.5s of posture change.

Camera distance variance: All metrics are angular or proportional (not pixel-based), so thresholds remain valid from 0.5m to 2.5m.

Lighting variability: MediaPipe handles varied lighting well; we advise front-facing light and avoid backlit windows.

6. How This Project Meets Grading Criteria

Criterion	Weight	How Addressed
Uniqueness	30%	EMA-smoothed multi-metric biomechanical posture scoring. 4 simultaneous body metrics. Session analytics.
Creativity	25%	Full web dashboard: live skeleton overlay, animated ring, score chart, AI coaching tips via Anthropic API.
Usefulness	20%	1B+ desk workers globally. Free, instant, private alternative to costly assessments or hardware.
Working Demo	15%	python app.py → browser → click Start. Works on any webcam. Graceful fallback if camera unavailable.
Code Quality	10%	Modular PostureAnalyzer class, docstrings, separation of inference/server/frontend.

7. Running the Project

Web Dashboard (recommended):

```
pip install -r requirements.txt
python app.py # Open http://localhost:5000
```

CLI inference:

```
python inference.py --source 0 # Webcam
python inference.py --source video.mp4 # Video file
python inference.py --source image.jpg # Single image
```